



CJC

Complete Java Classes

by Kunal Sir

What are Variables?

A variable is the name of a reserved area allocated in memory.

It is a combination of “**vary** + **able**” which means its value can be changed.

Variables in Java: -

A variable is a container which holds the value while the java program is executed.

A variable is assigned with a data type.

Variable is a name of memory location.

Int age = 20;

Data Variable_name Value
Type



RAM



How can we declare a variable: -

Datatype variable_name; // Declaration of variable

Example: - int x;

How can we initialize a variable: -

Variable_name = value; // Initialization of variable

Example: - x = 10;

How can we declare and initialize a variable:

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Datatype variable_name = value; // Declaration and Initialization of variable

Example: - int x = 10;

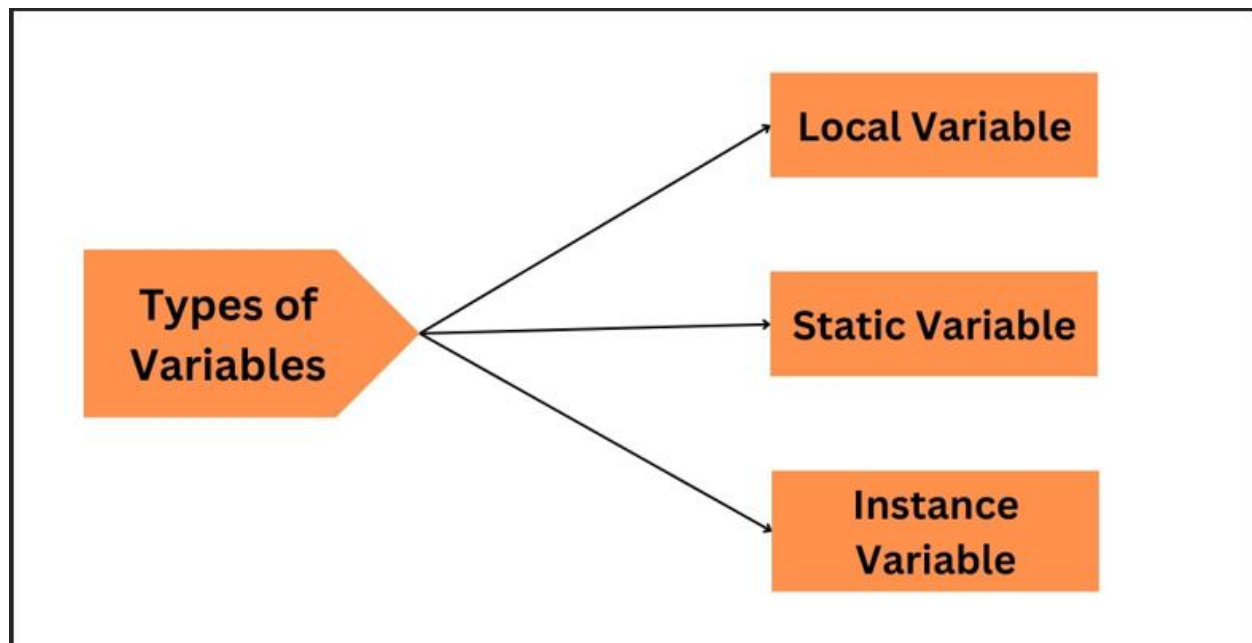
Types of Java Variables: -

There are three types of variables in java which is given



below -

1. Instance Variable/Non-static variable
2. Static Variable/Class Variables
3. Local Variable



Instance Variable/Non-static variable: -

A variable declared inside the class but outside the body of the method, is called an instance variable. It is not declared as static.



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Member variable and fields are other names for instance variable.

These are connected to the production of objects. Instance variable is also created as the object is formed.

This one is a part of global variable in java.

Static Variable/ Class Variable: -

Class variables also known as static variables are declared with the static keyword in a class, but outside a method, constructor or a block.

When a class is loaded into the JVM, they are loaded and initialized.

The class variables only exist once.

These variables have an implicit default value assigned to them if they are not initialized.



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These create when the program starts. It gets destroyed when the program stops.

This one is also part of global variable in java.

Local Variable: -

Local variable is declared in methods, constructor & block.

Local variables are created when the method, constructor or block is entered and the variable will be destroyed once it exits the method, constructor or block.

Access Modifier cannot be used for local variables.

There is no default value for local variables, so local variables should be declared and an initial value should be assigned before the first use.

Difference Between Global and Local Variables in Java:

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There are some differences between the local and the global variables which are given below –

Global Variable	Local Variable
Outside of every function block, global variables are declared.	The declaration of local variables occurs within a function block.
The scope remains throughout the program.	The scope is limited and remain within the function only in which they are declared.
We can not declare many variables with the same name.	We can declare various variables with the same name but in other functions.
If the global variable is not initialized, it takes zero by default.	If the local variable is not initialized, it takes the garbage value by default.
A global variable exists in the program for the entire time the program is executed.	A local variable is created when the function is executed, and once the execution is finished, the variables is destroyed.
Global variables are stored in the data segment of memory.	Local variables are stored in a stack in memory.



How we can write the different data types variables in java: -

In java, there are different types of variables, for an example-

1. **String** → Stores text, such as "CJC". String values are surrounded by double quotes (" ").
2. **int** → Stores integers (whole numbers), without decimal, such as 123 or -123 like that.
3. **float** → Stores floating point numbers, with decimal, such as 19.99 or -19.99 like that.
4. **char** → Stores single character, such as 'a' or 'A'. Char values are surrounded by single quotes (' ').
5. **boolean** → Stores values with two states: true or false.